Chapter 2 & 8

Layer Architecture

Layer	Core Function	Typical Protocols
Application Layer	Provide user-visible network services	HTTP, FTP, DNS, DHCP
Transport Layer	End-to-end communication, reliability, flow control	TCP, UDP
Internet Layer	Logical addressing, routing across networks	IP, ICMP, ARP
Link Layer	Data transmission within same network segment	Ethernet, Wi-Fi, PPP
Physical Layer	Actual physical signals & medium transmission	Cables, Signals, Hubs

IPv4 Packet Header

Field	Notes
Version	Version number (IPv4 = 4)
IHL	Header length (unit: 4 bytes)
Type of Service	Service type
Total Length	Max 65535 B
Identification	Identifier
Flags / Fragment Offset	Flags / fragment offset — ”don't / more fragment”
Time to Live (TTL)	Decremented by 1 per hop
Protocol	upper layer — TCP(6), UDP(17), ICMP(1)
Header Checksum	Header checksum
Source IP Address	Source IP address
Destination IP Address	Destination IP address
Options / Padding	Options / padding (align to 4-byte boundary)

IPv6 Basic Header

Field (offset)	Description	Notes
0-3	Version	IPv6 = 6
4-7	Traffic Class	Equivalent to IPv4 TOS
8-19	Flow Label	Marks data flow
20-31	Payload Length	Data length excluding header
32-39	Next Header	Identifies upper layer protocol or extension header
40-47	Hop Limit	Equivalent to IPv4 TTL, max 255
48-175	Source Address	128-bit IPv6 address
176-303	Destination Address	128-bit IPv6 address

IP Header Processing Steps

1. Compute header checksum, check version number and total length
2. Determine next hop from routing table
3. Modify TTL and recompute checksum

IP Address Classification (Classful Addressing)

Address Format:

Class	Format	Range	Default Mask	Purpose	Max Hosts
A	0xxx	1.0.0.0 – 126.255.255.255	255.0.0.0 /8	Huge Organizations	16 Million
B	10xx	128.0.0.0 – 191.255.255.255	255.255.0.0 /16	Large/Medium Orgs	65,534
C	110x	192.0.0.0 – 223.255.255.255	255.255.255.0 / 24	Small LANs	254
D	1110	224.0.0.0 – 239.255.255.255	N/A	Multicast	N/A
E	1111	240.0.0.0 – 255.255.255.255	N/A	Reserved/ Experimental	N/A

Figure 1: classful IP

Address Range & Capacity:

Class	Range	Format	Networks	Hosts
A	0–127	N.H.H.H	2^7	$2^{24} - 2$
B	128–191	N.N.H.H	2^{14}	$2^{16} - 2$
C	192–223	N.N.N.H	2^{21}	$2^8 - 2$

Private Addresses:

- Class A: 10.0.0.0 – 10.255.255.255
- Class B: 172.16.0.0 – 172.31.255.255
- Class C: 192.168.0.0 – 192.168.255.255
- Others are public addresses

NAT (Network Address Translation):

- Function: Translates between private ↔ public addresses
- Principle: Uses many port numbers; address mapping at gateway

Subnet Mask Table

Prefix	Mask	Prefix	Mask
/1	128.0.0.0	/5	248.0.0.0
/2	192.0.0.0	/6	252.0.0.0
/3	224.0.0.0	/7	254.0.0.0
/4	240.0.0.0	/8	255.0.0.0

IP communication: encapsulation

1. An IP packet contains source/destination IP addresses and the data from TCP/UDP. Then IP address (or the next-hop router's IP) -ARP-⟶ the physical MAC address.
2. Create an Ethernet frame containing payload and header. The IP packet becomes the payload of the Ethernet frame. The header contains the source and destination MAC addresses.
3. This Ethernet frame is converted into electrical (or radio) signals on the wire.
4. The receiving device's Ethernet card reads the frame, strips off the Ethernet header, sees the packet is for its IP address, and passes the IP packet up to its IP layer software.

Headers in TCP/IP communication?

Headers are small blocks of data added at the beginning of each layer's payload, containing the control information needed for that layer's protocol to do its job.

- **IP Header**: Contains source/destination IP addresses, time-to-live (TTL), protocol type (TCP=6, UDP=17), etc.
- **TCP Header**: Contains source/destination port numbers, sequence & acknowledgment numbers for reliability, flags (SYN, ACK, FIN), and window size for flow control.
- **Ethernet Header**: Contains source/destination MAC addresses.

TCP/IP Encapsulation

Ethernet h	IP h	TCP h	HTTP Request	IP tail	FCS

- **Ethernet h**:Data Link Layer — contains network type information
- **IP h**: Network Layer — contains IP address information
- **TCP h**: Transport Layer — contains port numbers (source, destination)
- **FCS**: Frame Check Sequence — error detection

CIDR (Classless Inter-Domain Routing)

- Core idea: Route based on address prefix (mask) — no strict A/B/C classes
- Example: `192.168.1.0/24` — first 24 bits are network number

Mobile IP

- Core concepts: Corresponding host, Home agent, Foreign agent
- Core mechanism: IP-in-IP encapsulation — transparent routing for mobile nodes

Autonomous Systems & Routing Protocols

Autonomous System (AS): Also called routing domain

- **Stub AS**: Only one connection to outside
- **Multihomed AS**: multiple external connections
- **Transit AS**: multiple external connections; can forward traffic

ICMP (Internet Control Message Protocol):

- Ping — test connectivity
- Traceroute — trace route

Chapter 5

Adaptation Functions (App ↔ Network)

- message size
- reliability, in-order delivery
- pacing & flow control
- timing for voice, audio, video
- addressing

Flow Control

- On-off: Receiver sends on or off signals
- Sliding Window: receiver always remain $2t_{prop}R$ buffer

ARQ

- Flow control & reliable tx
- Method: Retx
- Packets delivered in order; errors/duplications handled

Error Control in

- **Layer4** (TCP): end to end error check, retx
- **Layer3**: segments lost /order x /path
- **Layer2**: operates on directly-connected systems

Stop-and-Wait ARQ

1. Frames and ACKs must have sequence numbers
2. Sender state ~ last frame sent
3. Receiver state ~ next expected frame
4. $T_0 = 2t_{prop} + 2t_{proc} + t_{fr} + t_{ack}$ where $t_{fr} = n_f/R$ and $t_{ack} = n_a/R$

Error-Free Channel Efficiency

$$R_{eff} = \frac{N_f - N_o}{T_0}$$
$$\eta_{sw} = \frac{R_{eff}}{R} = \frac{N_f - N_o}{T_0 \cdot R}$$

Channel with Errors Efficiency

$$R_{eff} = \frac{N_f - N_o}{T_0/(1 - P_f)}$$
$$\eta_{sw} = \frac{R_{eff}}{R} = \frac{1 - \frac{N_o}{N_f}}{1 + \frac{N_a}{N_f} + \frac{2(t_{prop}+t_{proc})R}{N_f}} \cdot (1 - P_f)$$

Delay Component in Stop-and-Wait

$$E[Delay] = (1 - P_f)T_0 + P_f(t_{out} + E[Delay])$$
$$\Rightarrow E[Delay] = \frac{T_0}{1 - P_f} \quad \text{(assuming } t_{out} = T_0\text{)}$$
$$\text{Efficiency} = \frac{N_f - N_o}{E[Delay] \cdot R}$$

Go-Back-N ARQ

- Use m-bit sequence numbering
- Must ensure $N < 2^m = M$ (otherwise receiver confuse)
- ACK x: expect x next
- NAK x: expect x but not received
- Out-of-sequence frames are **discarded**

Selective Repeat ARQ

Receiver

- accepts/buffers out-of-sequence frames
- ACK can be advanced
- Send NAK with seq# R_{next} when out-of-sequence frame received

Sender

- Timer expires → retransmit only expired frame
- NAK i received → retransmit only frame i

To save space, we ensure
$$2N \leq 2^m = M$$
TCP: Variation of selective repeat but M must be larger than N since TCP × FIFO
Note: P_f = probability frame arrives without error, N = window size, m = bits for sequence number

Data Link Control

- Framing
- High-level Data Link Control (HDLC)
- Point-to-point Protocol (PPP)
- Link Sharing Using Statistical Multiplexing

Framing - Purpose

- Define boundaries of active/idle
- Preamble for synchronizing clocks

HDLC: Bit-oriented

- **Normal Response Mode (unbalanced)**: P2P link, multipoint link
- **Asynchronous Balanced Mode (balanced)**

Frame Structure

Flag	Address	Control	Info	FCS	Flag
①	②	③	Data	Frame Check Seq	①

① **Flag**

- **01111110**: indicates message boundaries
- **11111111** → Idle
- **01111111** → Abort (followed by Idle)

Bit-stuffing

- **Purpose**: prevent flag pattern from occurring inside info
- **Rule**: Transmitter inserts 0 after each instance of 5 consecutive 1's
- Receivers do the reverse

② **Address**: distinguishes between secondaries in unbalanced mode

③ **Control Field (1 byte)**

(1) Information Frame (I-frame)

0	N(S)	P/F	N(R)
1	2-4	5	6-8

- N(S): Sequence Number (transmit)
- P/F: Poll/Final — unbalanced mode: polling; balanced mode: checkpointing
- N(R): Piggybacked ACK (next expected frame)

(2) Supervisory Frame (S-frame)

1	0	S	S	P/F	N(R)

Types of SS: (for flow control)

- 00: Receive Ready (RR)
- 01: Reject (REJ) — NAK of GBN
- 10: Receive Not Ready (RNR)
- 11: Selective Reject (SREJ) — NAK of SR

(3) Unnumbered Frame (U-frame)

1 1	M	M	P/F	M	M	M

M bits express:

- SABM (Set Asynchronous Balanced Mode)
- SNRM (Set Normal Response Mode)
- SABME (Extended)
- DISC (Disconnect)
- UA (Unnumbered Ack)
- FRMR (Frame Reject)

HDLC do flow control by

- Receivers delaying acknowledge messages
- Receivers use supervisory frames to control flow

Point-to-Point Protocol (PPP: Byte-oriented)

Functions:

1. Provide Framing & Error Detection
2. Link Control Protocol (connection-oriented)
3. does not implement ARQ, leaving error recovery to higher layers (TCP)
4. simpler, build direct connections between two nodes
5. provides features for authentication (PAP/CHAP) and link quality testing

Frame Structure

Flag	Address	Control	Protocol	Info	CRC	Flag
01111110	11111111	00000011				01111110

- **Address**: no meaning — PPP uses unnumbered frames
- **Control**: unnumbered frame
- **Protocol**: can support multiple network protocols simultaneously
- **Information**: has integer number of bytes

Byte-stuffing

- **Rule**: Any occurrence of flag (0x7E) or control escape (0x7D) inside frame is replaced with 0x7D followed by original octet XORed with 0x20 (00100000)

Timing Recovery (for synchronous services)

we use playout buffer:

1. Delay first packet by maximum network delay
2. All other packets arrive with less delay
3. Playout packets uniformly thereafter

Chapter 6a

Basics

Term	Definition/Notes
LAN Overview	private ownership, short distance (≈1km), high speed, low bit error, broadcast transmission, no complex routing, frequently moved machines
LAN Structure	transmission medium, NIC (network interface card), unique MAC “physical” address
LLC (802.2)	Data Link Layer sublayer, protocol multiplexing + error control, 3 service types (Unacknowledged Connectionless is most used)
MAC Protocol	Data Link Layer sublayer, coordinates node medium sharing, reduces/avoids collisions
MAC Address	48-bit globally unique, burned into NIC ROM, identifies LAN nodes

Logical Link Control Services

- Unacknowledged connectionless service: Unnumbered frame mode of HDLC
- Reliable connection-oriented service: Asynchronous balanced mode of HDLC
- Acknowledged connectionless service: Two additional unnumbered frames added to HDLC

Key Performance Parameters

- $t_{\text{prop}} = d/v$ (propagation delay)
- # wasted bits = $2t_{\text{prop}} * R$ where R is the data transmission bit rate.
- $t_f = L/R$ (frame transmission time, L bits, R bit/s, $X := t_f$)
- $a = t_{\text{prop}}/t_f$ (normalized delay–bandwidth product, smaller a is better)
- Normalized Delay-Bandwidth Product (unit-less):

$$a = \frac{t_{\text{prop}}}{L/R} = \frac{t_{\text{prop}}}{t_f}$$

- Throughput S: actual successful rate (normalized ≤ 1)
- Max Throughput:

$$R_{\text{eff}} = \frac{L}{t_f + 2t_{\text{prop}}} = \frac{L}{L/R + 2t_{\text{prop}}} = \frac{1}{1+2a}R \quad \text{(bits/second)}$$

- Efficiency ρ: effective transmission / total channel occupancy
- Max Possible Efficiency (unit-less): (not achievable in practice)

$$\rho_{\max} = \frac{L}{L + 2t_{\text{prop}}R} = \frac{1}{1 + 2t_{\text{prop}}R/L} = \frac{1}{1+2a}$$

- Load G: attempts per t_f time unit (core for random access)

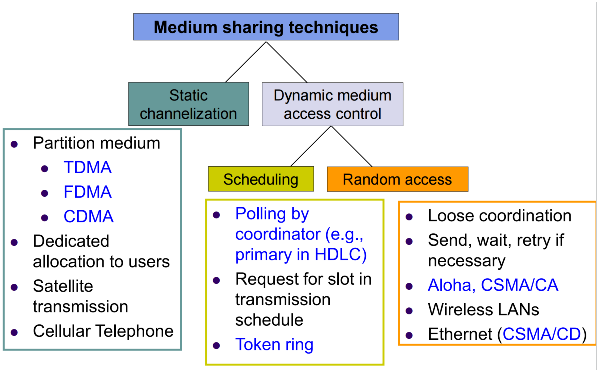


Figure 2: medium access control in Layer 2

Major Media Sharing Methods

Type	Static	Dynamic	Random
Description	Pre-divide channel	Unified scheduling	Transmit when ready
Pros	No collision, deterministic	High utilization, fairness, QoS	Simple, bursty, low cost
Cons	Resource waste	High complexity, overhead	High collision at load
Apps	Satellite/cellular	Industrial control	Ethernet/WiFi
Examples	TDMA / FDMA / CDMA	Polling / Token / Reservation	ALOHA / CSMA / CSMA-CD / CA

Random Access Protocols

ALOHA Series

$$E[\text{Delay}] = X + t_{\text{prop}} + \left(e^{2G} - 1\right)(X + 2t_{\text{prop}} + E[B])\ (s)$$

$$E[\text{Delay}] = 1 + a + \left(e^{2G} - 1\right)\left(1 + 2a + \frac{E[B]}{X}\right)(t_{frame})$$

Protocol	Vuln. Period	Throughput Formula	Max S	Sync
Pure ALOHA	$2X$	$S = Ge^{-2G}$	0.184 ($G = 0.5$)	No
Slotted ALOHA	X	$S = Ge^{-G} = P[1]$	0.368 ($G = 1$)	Yes

$$P[\text{0-trans}] = \frac{G^0}{0!}e^{-G} = e^{-G} = P_{\text{idle}}$$

$$P[\ge 2] = P[\text{collision}] = 1 - e^{-G} - Ge^{-G}$$

Slotted ALOHA halves vulnerable period, doubles throughput.

Random Access Core Mechanisms

- *Carrier Sensing*: Detect channel before tx, shortens vulnerable period.
- *Collision Detection (CD)*: Sense while tx, stop + jam if collision.
- *Collision Avoidance (CA)*: Wireless, RTS/CTS + pre-backoff.
- *Backoff*: Random wait; CSMA-CD uses **Binary Exponential Backoff**.

CSMA Series (Vulnerable = t_{prop})

Type	Algorithm	Performance
1-Persistent	Continuous sensing, tx when idle	Low delay, high collision at heavy load
Non-Persistent	Random backoff, re-sense	Low collision, high utilization, high delay
p-Persistent	Tx with prob p / wait t_{prop} with $1-p$	Balance delay & utilization (WiFi basis)

CSMA/CD (Ethernet 802.3, wired)

Carrier Sensing → Transmission → Collision Detection → Backoff & Retransmit.
Collision detection time: $2t_{\text{prop}}$ (Ethernet requires $t_f > 2t_{\text{prop}}$, i.e. $a < 0.5$).
Efficiency: $\rho = \dfrac{1}{1+6.44a}$ $\quad(\rho \to 100\%$ when $a \to 0)$.
More resources will be wasted if the delay-bandwidth product is higher.

CSMA/CA (WiFi 802.11, wireless)

Collision *avoidance* (RTS/CTS + random backoff after idle). Adapts to wireless, lower utilization than CD but no collision.

Dynamic Scheduling Protocols

Token Ring (802.5): Token circulates → capture (busy) → tx → restore token.
Efficiency: $\rho = \dfrac{1}{1+a'}$ (a' = ring latency (bit) / avg frame length).
No collision, deterministic delay; single point of failure.
Polling/Reservation: Central controller polls / nodes reserve first. Suits QoS.

MAC Delay Performance

Frame transfer delay: From first bit at source to last bit at destination.

- If no waiting we have minimized delay as $t_f + t_{\text{prop}}$
- Min. normalized: $\dfrac{t_f + t_{\text{prop}}}{t_f} = 1 + a$

Throughput: Actual transfer rate (frames/s or bits/s).

- R bits/s, L bits/frame
- $X = L/R$ s/frame (frame duration)
- λ frames/s (arrival rate)
- Load $\rho = \lambda X$ (unit-less)
- Max throughput: $R/L = 1/X$ frames/s

Comparisons

- *Max Throughput*: CSMA/CD > CSMA > Slotted ALOHA > Pure ALOHA
- *Media*: wired (CSMA/CD, Token Ring), wireless (ALOHA, CSMA/CA)
- *Optimize* a: ↑ min frame length L (Ethernet 64B) or ↓ d or ↓ R
- IEEE 802.2: LLC enhances service provided by MAC with (1) Multiplexing of different network protocols; (2) Error control
- *IEEE 802*: 802.3 (Ethernet), 802.5 (Token Ring), 802.11 (WiFi)
- Random access: S ↑ with G at light load; collision surges when G exceeds threshold
- *Binary Exponential Backoff*: backoff range doubles with each collision
- *Ethernet key requirement*: $X > 2t_{\text{prop}}$ ensures collision detection before end of frame

Chapter 6b

1. 802.3 vs 802.11

802.3: Wired Ethernet standard.
802.11: Wireless LAN (Wi-Fi) standard.

2. Ethernet Frame Types

Three historic formats; today mostly **Ethernet II**:

- **Ethernet II (DIX)**: Has Type field (e.g., IP). Most common.
- **IEEE 802.2/802.3 (LLC)**: Length field + LLC header; complex.
- **Novell 802.3 raw**: Rare.

Analogy: Ethernet II envelope says "what's inside"; 802.3 envelope says "weight" plus inner label.

3. Repeater, Hub, Bridge, Switch, Router

- **Repeater** (L1): Amplifies signal; no addressing.
- **Hub** (L1): Multi-port repeater; broadcasts to all.
- **Bridge** (L2): Learns MAC addresses; 2 ports; filters/forwards.
- **Switch** (L2): Multi-port bridge; core of modern LANs.
- **Router** (L3): Uses IP addresses; connects different networks.

Dimension	Repeater/Hub	Bridge/Switch	Router
Core Mechanism	Signal (L1)	MAC filtering (L2)	IP routing (L3)
Traffic	All segments see traffic	Local isolation	Cross-network
Collision domains	1	2 (each port)	2
Broadcast domains	1	1	2
MAC addresses	No	One mgmt MAC	Two MACs + IPs
MAC header change	No	No	Yes

4. How Bridges Learn Locations

Self-learning:

1. Start: empty MAC table.
2. On receiving frame, record **source MAC + port + timestamp**.
3. Forward/filter: if dest MAC known and port different, forward; if same port, drop; if unknown, flood.

5. Loop Prevention (STP)

Why prevent loops? Broadcast storms, MAC table instability.
Solution: Spanning Tree Protocol (STP).

- Elect one **Root Bridge**.
- Each switch selects **Root Port** (best path to root).
- Each segment selects **Designated Port**.
- Remaining ports are **blocked** (logical break).

6. WLAN Operation

Components: Station (phone/PC) and Access Point (AP).
AP sends **beacons**; station scans, authenticates, associates. Data travels via AP.

7. BSS and ESS

- **BSS**: Basic Service Set — one AP + its stations.
- **BSA**: Basic Service Area (coverage).
- **ESS**: Extended Service Set — multiple BSSs with same SSID, connected via distribution system (wired).

8. CSMA/CA vs CSMA/CD

CSMA/CD (wired): Listen while talking; detect collision; stop; random backoff.
CSMA/CA (wireless): Listen before talk; avoid collision (cannot listen while sending due to strong self-signal).

Why not CD in WLAN? Transmitter deafens its own receiver; hidden terminal problem.

9. Hidden & Exposed Terminals

Hidden: A and B both see AP but not each other → collide at AP.
Solution: RTS/CTS handshake.

- A sends RTS (Request to Send) to AP.
- AP replies CTS (Clear to Send); B hears CTS, defers.

Exposed: C hears A transmitting to B, but C could send to D (different destination); C defers unnecessarily. Partially solved by RTS/CTS.

10. RTS and CTS

Optional mechanism to reserve channel and solve hidden terminal.

- **RTS**: small frame from sender: "I need to send".
- **CTS**: from receiver: "OK, everyone be quiet for duration".
- Duration field sets NAV on listening stations.

11. PCF vs DCF

DCF (Distributed Coordination Function): CSMA/CA, contention-based, default.
PCF (Point Coordination Function): AP polls stations, contention-free, optional (rare).

12. Virtual Carrier Sense & NAV

NAV (Network Allocation Vector): counter in each station; set by duration field in RTS/CTS/data frames. While NAV > 0, station considers channel busy *virtually* — even if physical carrier is idle.

13. Ad Hoc vs Infrastructure

Infrastructure: uses AP; all communication via AP; connects to Internet.
Ad Hoc (IBSS): no AP; stations talk directly; temporary, peer-to-peer (e.g., file transfer).

14. SIFS, PIFS, DIFS

Inter-frame spaces (shorter = higher priority):

- **SIFS**: shortest — for ACK, CTS (immediate response).
- **PIFS**: medium — for PCF polling (AP use).
- **DIFS**: longest — for regular data/RTS in DCF.

15. Quick Reference

802.3 (Ethernet): CSMA/CD, switches, frames.
802.11 (Wi-Fi): CSMA/CA, APs, RTS/CTS, NAV.

16. Statistical Multiplexing

Self Delay = Waiting + Service Times

Hop-by-hop (HDLC, PPP, WiFi):

- Pros: Fast error recovery, prevents error propagation
- Cons: Overhead on every hop, buffering need

End-to-end (TCP):

- Pros: Simpler, only endpoints have complex logic
- Cons: Slow recovery, duplicates low-layer effort

Chapter 7a

1. Network Layer Functions

Essential:

- **Routing**: build forwarding table collaboratively
- **Forwarding**: based on routing table
- **Priority & Scheduling**: determining order of packet transmission

Optional: congestion control, segmentation & reassembly, security.

2. DNS vs. ARP

Feature	DNS	ARP
Layer	Application Layer (5)	Data Link Layer (2)
Purpose	domain name → IP	IP → MAC
Query Type	Unicast to DNS server	Broadcast on subnet
Response Type	Query Response	Request Response
Config	DNS server address got by DHCP or manual	No configuration required; built into network stack

Discovery Process Overview

1. **IP Address Discovery (DNS):** (please review the process if necessary)
2. **MAC Address Discovery (ARP):**
   - **Same subnet:** PC broadcasts an ARP request; destination replies with its MAC address
   - **Different subnets:** PC uses ARP to find the MAC address of its default gateway (router), not the final destination

3. Packet vs. Message Switching:

Feature	Message Switching	Packet Switching
Unit of Data	large, variable size	typically < 1.5KB
Storage	store the entire message	store small packets
Delay	High latency	Low latency
real-time	unsuitable	suitable
Resource Usage	Ties up the link until the entire message is txed	Allows multiple users share links
Error Handling	the entire message must be resent	only that small packet resent

Store-and-forward

Reasons: delay (pipelining), loss (retransmit less), fairness.

Delay (L hops, k packets, T/k, τ(propagation delay)):

Message: $L\tau + LT$

Packet:

First bit $L\tau + (L-1)\frac{T}{k}$

Last bit $L\tau + \frac{LT}{k} + (k-1)\frac{T}{k}$

Last bit $T_t = \frac{T}{k}, \quad L\tau + (k+1)T_t$ if L=2

4. Packet Switch Architecture

Line cards (PHY+DL+Network) → interconnection fabric → control.

Examples: PC router, WLAN AP.

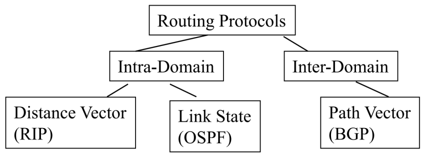
5. Datagram vs. Virtual Circuit

Datagram: full IP in every packet, independent routing.

Virtual Circuit: setup + short VC ID, same path, tables.

Aspect	Datagram	Virtual Circuit
Towards	connection-less	connection-oriented
Each packet	has full source/destination	has short VCI / VPI
Routing	dynamic, independent	fixed for each connection
Routing Table	large and hierarchical	small
Overhead	Higher (full address)	Lower (small VC label)
Packet Order	may arrive out-of-order	arrive in-order
Reliability	Best-effort; no guarantee	guarantee QoS
Failure Recovery	Easy to reroute	Must establish new VC
Examples	IP, Ethernet, UDP	ATM, MPLS, X.25
Resource Reserve	Difficult; best-effort	Possible via CAC at setup
Scalability	Highly scalable	Limited by per-flow state

6. Families of routing protocols



Concept	Distance-Vector Routing	Link-State Routing
How it works	"Routing by rumor"	"Global mapping"
	tell neighbors	tell everyone
	Routers share their entire routing table only with their immediate neighbors at regular intervals	Each router "floods" information about its directly connected links to every router in the area
Metrics	Uses a simple metric (hop count)	More sophisticated (cost, bandwidth, delay, load)
Convergence	Slow. Easy to get loops	Very fast. Less prone to loops.
Algorithm	Bellman-Ford	Dijkstra's Shortest Path First (SPF)
Examples	RIP, IGRP	OSPF, IS-IS

7. Routing Algorithms

Distance-Vector (Routing Infomation Protocol): Bellman-Ford.

$$D_i = \min_j (C_{ij} + D_j)$$

Link-State (Open Shortest Path First): flood topology → each runs Dijkstra.

Special Routing Strategies:

- Flooding (TTL control)
- Source routing (full path in header)
- Deflection routing (busy → deflect, "Hot Potato" or "Bufferless Routing")

8. Loss Example

$p = 10^{-6}$, $L = 10^6$ bits → whole message $\approx e^{-1} \approx 0.368$

$k = 10$ packets: each ≈ 0.9 → all succeed ≈ 0.368

Why short packets? Fewer bits retransmitted!

Summary

- Memorise delay graphs (pipelining)
- Review DNS and ARP process
- Draw datagram vs. VC tables
- Run Bellman-Ford & Dijkstra(let gravity do the work) examples

Chapter 7b

ATM Networks

Key Features

- **Asynchronous Transfer Mode**: 1980s telecom standard.
- **Fixed-size cells**: 53 bytes (48 payload + 5 header) → fast hardware switching.
- **Connection-oriented**: Virtual circuits (VPI/VCI) with QoS negotiation.
- **QoS classes**: CBR, VBR, ABR, UBR.

Why ATM Declined

- Complex (SAR, signaling) vs. simple Ethernet/IP.
- High overhead (≈10% cell tax).
- Expensive, low volume vs. Ethernet economies of scale.
- IP became dominant.

ATM vs. IP

Feature	ATM	IP/Ethernet
Packet Size	53-byte cells	64 - 1500+ bytes
Connection	Connection-oriented (virtual circuit established first)	Connectionless (each packet is routed independently)
Switching	Hardware-based (fast)	Software-based (slower)
QoS	Built-in, robust QoS with guaranteed classes of service (CBR, VBR, ABR, UBR)	Best-effort, QoS complex to implement (e.g., DiffServ, MPLS)
Complexity	Complex edge, simple core	Simple edge, complex core
Overhead	High (10% cell tax)	Lower for large data

Traffic Management & Congestion Control

Traffic Management Packet Level: Queuing

Packet-by-packet model delivers time of the last packet is the same as the fluid model, but for the previous packets, it arrives earlier than the fluid model.

FIFO, Priority, Earliest Due Date Queuing, Fair Queuing, Weighted Fair Queuing.

Congestion: offered load > capacity → delay, loss.

Open-Loop vs Closed-Loop (flow level)

Aspect	Open-Loop Control	Closed-Loop Control
Approaches	Preventive	Reactive
Feedback	No feedback; decisions be made before flow starts	Uses explicit/implicit feedback; continuous adjustment
Guarantees	hard QoS guarantees	soft best-effort
Efficiency	Low for bursty traffic	High for bursty data traffic
Stability	stable	oscillate possible due to feedback delay
Scalability	Limited (per-flow state)	High (aggregate state, no per-flow reservations)
Complexity	Complex admission control	Complex algorithms at end-points (e.g., TCP)
Best for	Real-time, CBR, streaming	Data, bursty, elastic traffic
Examples	ATM, RSVP, telephone network	TCP, ATM ABR

Open-Loop Control Examples

- **Admission control**(connection level): CAC reserves resources (peak rate, burst size).
- **Policing**(packet level): Leaky Bucket – drops/tags non-conforming cells.
- **Shaping**(packet level): Token Bucket smooths traffic.

Summary

TCP: Reactive closed-loop control (AIMD, slow start, fast retransmit/recovery) dynamically adjusts cwnd based on RTT/loss.
ATM: Open-loop with admission control and fixed cells; lost to IP due to complexity and cost.

TCP Congestion Control Fundamentals

RTT and Congestion Window

$$\text{Throughput} \approx \frac{\text{cwnd}}{\text{RTT}}$$

BDP = Bandwidth × RTT. Window ≥ BDP to fully utilize link.

TCP Windows

- **cwnd**: sender's estimate of network capacity.
- **awnd**: receiver's buffer space (flow control).
- **Actual window** = $\min(\text{cwnd}, \text{awnd})$.

TCP Algorithms

Slow Start

- Start: cwnd = 1 MSS.
- On each ACK: cwnd += 1 (exponential: doubles per RTT).
- Continues until loss or **ssthresh** reached.

Congestion Avoidance

- When cwnd ≥ ssthresh: linear increase (cwnd += 1/cwnd per ACK).
- AIMD: Additive Increase Multiplicative Decrease.

Fast Retransmit

- 3 duplicate ACKs → packet loss inferred.
- Retransmit without waiting for timeout.

Why TCP Doesn't Use NAK

- NAK used at Data Link Layer or when out-of-order delivery impossible
- NAK N implies packet N definitely lost (triggered by later packet)
- TCP uses IP — packets can travel different paths
- Then out-of-sequence delivery very possible in TCP connections

Fast Recovery

- After fast retransmit: ssthresh = cwnd/2; cwnd = ssthresh + 3.
- For each additional dup ACK: cwnd += 1.
- On new ACK: cwnd = ssthresh (exit recovery).
- Avoids going back to Slow Start for single loss.

Timeout

- Severe congestion indicator.
- ssthresh = cwnd/2; cwnd = 1; re-enter Slow Start.

RTT Estimation & RTO Formulas

Time from packet send to ACK receipt. Timeout adapts to network conditions.

Formula	Description
$\text{RTT}_{\text{smooth}} = \alpha \cdot \text{RTT}_{\text{old}} + (1-\alpha) \cdot M$	Smoothed RTT; $\alpha = 7/8$
$\text{DEV} = \delta \cdot \text{DEV}_{\text{old}} + (1-\delta) \cdot \lvert M - \text{RTT}_{\text{smooth}} \rvert$	Deviation estimate; $\delta = 3/4$
$\text{RTO} = \text{RTT}_{\text{smooth}} + 4 \cdot \text{DEV}$	Retx Timeout; adaptive

TCP Connection Management

Three-Way Handshake

1. **SYN**: Client → Server (seq = client_isn).
2. **SYN-ACK**: Server → Client (ack = client_isn+1, seq = server_isn).
3. **ACK**: Client → Server (ack = server_isn+1).

Establishes initial sequence numbers; enters ESTABLISHED.

Important Concepts

Terms	Definitions
Flow-level management	Per-connection decisions (admission control, policing, shaping)
BDP	Bandwidth × RTT; minimum window to fully utilize link
Leaky Bucket (Policing)	Constant leak rate; depth limits burst; non-conforming packets dropped/tagged
Token Bucket (Shaping)	Tokens accumulate at rate R; burst up to bucket depth; smooths traffic
ssthresh	Threshold between Slow Start and Congestion Avoidance
Duplicate ACKs	Signal out-of-order/loss; trigger fast retransmit

TCP Summary

Mechanism	Purpose
Advertised Window (awnd)	Flow control: prevents sender from overwhelming receiver's buffer
Timeout	Safety net: detects severe congestion or total ACK loss; triggers drastic reduction (cwnd=1)
Congestion Window (cwnd)	Congestion control: dynamically limits sender's data in flight to prevent network overload
Slow Start	Quickly ramps up transmission rate from a low baseline to find available bandwidth
Slow-Start Threshold (ssthresh)	Marks the transition from slow start to congestion avoidance to prevent overshooting network capacity
Fast Retransmit	Avoids timeout by retransmitting a lost packet after three duplicate ACKs, signaling moderate congestion
Fast Recovery	Maintains higher throughput after fast retransmit by setting cwnd to ssthresh and using AIMD